

# PROJECT REPORT

## UPI Payment QR Code Generator Using Python

A Python-based application for generating UPI payment QR codes

**Project Type:** Python / Digital Payment Application

**Development Environment:** Jupyter Notebook

## 1. Introduction

Digital payments have become an important part of modern financial transactions. UPI (Unified Payments Interface) allows users to make fast and convenient payments using a UPI ID or QR code. This project, "UPI Payment QR Code Generator Using Python," generates UPI payment QR-code images from a user-provided UPI ID.

## 2. Problem Statement

Manually entering payment details can be inconvenient and may result in incorrect information. The project provides a simple Python-based method for converting a UPI ID into a scannable QR code, making the sharing of payment information easier.

## 3. Objectives

- Generate UPI payment QR codes automatically.
- Accept a UPI ID as user input.
- Create a UPI payment URI.
- Convert the URI into a QR code.
- Save generated QR codes as image files.
- Demonstrate a practical application of Python in digital payments.

## 4. Technologies Used

The project uses Python as the main programming language and Jupyter Notebook as the development environment. The **qrcode** library is used to generate QR-code images, with Pillow support for image handling.

## 5. Libraries Used

The main library is the Python **qrcode** package. It can be installed using:

```
!pip install qrcode[pil]
```

The library is imported with:

```
import qrcode
```

## 6. Working of the Project

The project follows a simple workflow: the user enters a UPI ID; a UPI payment URL is created; the URL is encoded into a QR code; and the QR code is saved as a PNG image. The generated image can then be shared or scanned using a compatible UPI application.

## 7. Implementation

A typical implementation accepts the UPI ID using Python input and constructs a payment URI. For example:

```
upi_id = input("Enter your UPI ID = ")
```

```
phonepe_url = f'upi://pay?pa={upi_id}&pn=Recipient%20Name&mc=1234'
```

The QR image can then be created and saved using:

```
phonepe_qr = qrcode.make(phonepe_url)
phonepe_qr.save('phonepe_qr.png')
```

## 8. Output

The project produces QR-code image files such as **phonepe\_qr.png**, **paytm\_qr.png**, and **google\_pay\_qr.png**, depending on the implementation in the notebook. These files contain encoded UPI payment information.

## 9. Advantages

- Simple and easy to use.
- Reduces manual entry of payment information.
- Automatically generates QR-code images.
- Easy to save and share.
- Useful for shops, freelancers, demonstrations, and educational projects.

## 10. Limitations

The current implementation uses predefined recipient information and does not process or confirm transactions. It also does not provide a graphical user interface or payment gateway. Correctness of the generated QR code depends on the accuracy of the supplied UPI details.

## 11. Future Scope

The project can be extended with a graphical user interface, dynamic payment amounts, recipient-name fields, UPI-ID validation, customizable QR-code designs, download/share functionality, transaction history, database integration, and a web or mobile application.

## 12. Applications

The system can be used by small businesses, retailers, freelancers, individuals, donation campaigns, educational demonstrations, and as a foundation for larger digital-payment applications.

## 13. Conclusion

The UPI Payment QR Code Generator Using Python successfully demonstrates how QR-code technology can be combined with UPI payment information. By accepting a UPI ID, creating a UPI payment URI, and converting it into a QR code, the project provides a simple and practical digital-payment utility.

## 14. Project Outcome

The final outcome is a Python-based system capable of generating QR-code images from UPI payment information. The generated QR codes can be used with compatible UPI applications to simplify the entry of payment details.

## 15. References

- Python documentation
- Python qrcode library documentation
- UPI payment URI concepts
- Jupyter Notebook documentation